CS 702 - Final Term 2015

1. Steps of greedy algorithm design
2. Pseudo code of Huffman coding
3. Prove that circuit-sat is hard
4. Prove gcd(an,bn)=n.gcd(a,b)
5. Decrypted message (example in slides)
6. Backtracking to find maximum profit maximum weight is 8. see example in backtracking)
7. In DFS if vi is en-queued before vj then prove that d[vi]<d[vj] 5makrs
8. Bellman ford algorithm 5marks
9. Prove that a|0 for all a belong to Z 5 marks
10. Naïve String Matching Algorithm 5 marks
11. For all primes p and all integers a, b, if p | ab, then p | a or p | b (or p divides both and b)
12. Prime algorithm 10 marks
13. CNF-SAT is NP complete 10 marks
14. If X is problem such that $P' \leq pX$ for some $P' \in NPC$, then X is NP-hard. Moreover, $X \in NP \Rightarrow X \in NPC$ 15 marks
15. Write down pseudo code of longest common subsequence?
16. Write down pseudo code of Naive String Matching problem?
17. Write down pseudo code of optimal BST?
18. Write down pseudo code of transition function?
19. Write down prim's algorithm?
20. Write down Shortest Path (dag) algorithm?
21. If a > b ≥ 1 and the invocation EUCLID (a, b) takes k ≥ 1 recursive calls, then a ≥ Fk+2 and b ≥ Fk+1 ?
22. If p is prime, a is positive integer not divisible by p, ap-1 = 1 mod p OR ap = a mod p
23. If gcd(a, m) = 1 and m > 1, then a has a unique inverse a′ (modulo m).
24. If X is problem such that P'≤ p X for some P'∈NPC, then X is NP-hard. Moreover, X ∈ NP→X∈ NPC.
25. Why we use dynamic programming? Give limitations.....5 marks
26. Pseudo code for Johnson's algorithm....5 marks
27. How to print path in BFS algorithm...5 marks.
28. Prove that for a problem X and P' such that P' <p NPC , X is NP hard. ...10 marks
29. Extend Shortest Path algorithm.5 marks.

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